

and

$$[\mathbf{P}]_{j(x,y)} = \frac{1}{2\pi l_j^2} \begin{bmatrix} (x_{j+1} - x_j) & -(y_{j+1} - y_j) \end{bmatrix} \begin{bmatrix} [(l_j - \xi)\Phi + \eta\Psi] & (\xi\Phi - \eta\Psi) \\ [\eta\Phi - (l_j - \xi)\Psi - l_j] & (-\eta\Phi - \xi\Psi + l_j) \end{bmatrix} \quad (1.6.23)$$

The $n \times n$ airfoil coefficient matrix, $[\mathbf{A}]$, is generated from the 2×2 panel coefficient matrix in airfoil coordinates, $[\mathbf{P}]_{ji}$, for the velocity induced at control point i by panel j , extending from node j to node $j+1$, using the algorithm

$$A_{ij} = 0.0, \quad i=1, n; \quad j=1, n \quad (1.6.24)$$

$$\left. \begin{aligned} A_{ij} &= A_{ij} + \frac{x_{i+1} - x_i}{l_i} P_{21ji} - \frac{y_{i+1} - y_i}{l_i} P_{11ji} \\ A_{ij+1} &= A_{ij+1} + \frac{x_{i+1} - x_i}{l_i} P_{22ji} - \frac{y_{i+1} - y_i}{l_i} P_{12ji} \end{aligned} \right\}, \quad i=1, n-1; \quad j=1, n-1 \quad (1.6.25)$$

$$A_{n1} = 1.0 \quad (1.6.26)$$

$$A_{nn} = 1.0 \quad (1.6.27)$$

The n **nodal vortex strengths**, γ_1 through γ_n , are then obtained by numerically solving the $n \times n$ linear system

$$[\mathbf{A}] \begin{Bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_{n-1} \\ \gamma_n \end{Bmatrix} = V_\infty \begin{Bmatrix} [(y_2 - y_1)\cos\alpha - (x_2 - x_1)\sin\alpha]/l_1 \\ [(y_3 - y_2)\cos\alpha - (x_3 - x_2)\sin\alpha]/l_2 \\ \vdots \\ [(y_n - y_{n-1})\cos\alpha - (x_n - x_{n-1})\sin\alpha]/l_{n-1} \\ 0.0 \end{Bmatrix} \quad (1.6.28)$$

Once the nodal strengths are known, the velocity and pressure at any point in space can be computed by adding the velocity induced by all $n-1$ vortex panels to the free-stream velocity,

$$\begin{Bmatrix} V_x \\ V_y \end{Bmatrix} = V_\infty \begin{Bmatrix} \cos\alpha \\ \sin\alpha \end{Bmatrix} + \sum_{i=1}^{n-1} [\mathbf{P}]_{i(x,y)} \begin{Bmatrix} \gamma_i \\ \gamma_{i+1} \end{Bmatrix} \quad (1.6.29)$$

$$V^2 = V_x^2 + V_y^2 \quad (1.6.30)$$

$$C_p \equiv \frac{p - p_\infty}{\frac{1}{2}\rho V_\infty^2} = 1 - \frac{V^2}{V_\infty^2} \quad (1.6.31)$$